



What is Collection in Java?

- Collection it represents **group of objects**.
- In other words, a collection is group of individual data, that are grouped together to form a single unit.
Collection are containers that holds multiple data as a single unit.

For example: - During school time, you put all the important books in the school bag before going to school. Here schoolbag is **Collection** and books are **Objects**.

What is Framework in Java?

- Readymade interface, class and methods which is helpful for operation or manipulation.

What is Collection Framework in Java?



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- Collection framework is the **internal implementation of data structure**.
- It provides all the features of an array as well as overcome limitations of an array.
- In other words, Collection Framework in java an architecture using which we can represent a collection of interfaces and classes. It is used in storing, maintaining and handling data effectively.
- The Collection framework makes manipulations of the collection easier. This makes the task of a programmer simpler.

What is Data Structure?

- A data structure is a specialized format for organizing, processing, retrieving and storing data. It organizes data into different structures (basic and advance).
- Data structure makes it easy for users to access and work with the data they need in appropriate ways.
- Most importantly, data structures frame the organization of



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information so that machines and humans can better understand it.

Need of Collection Framework:

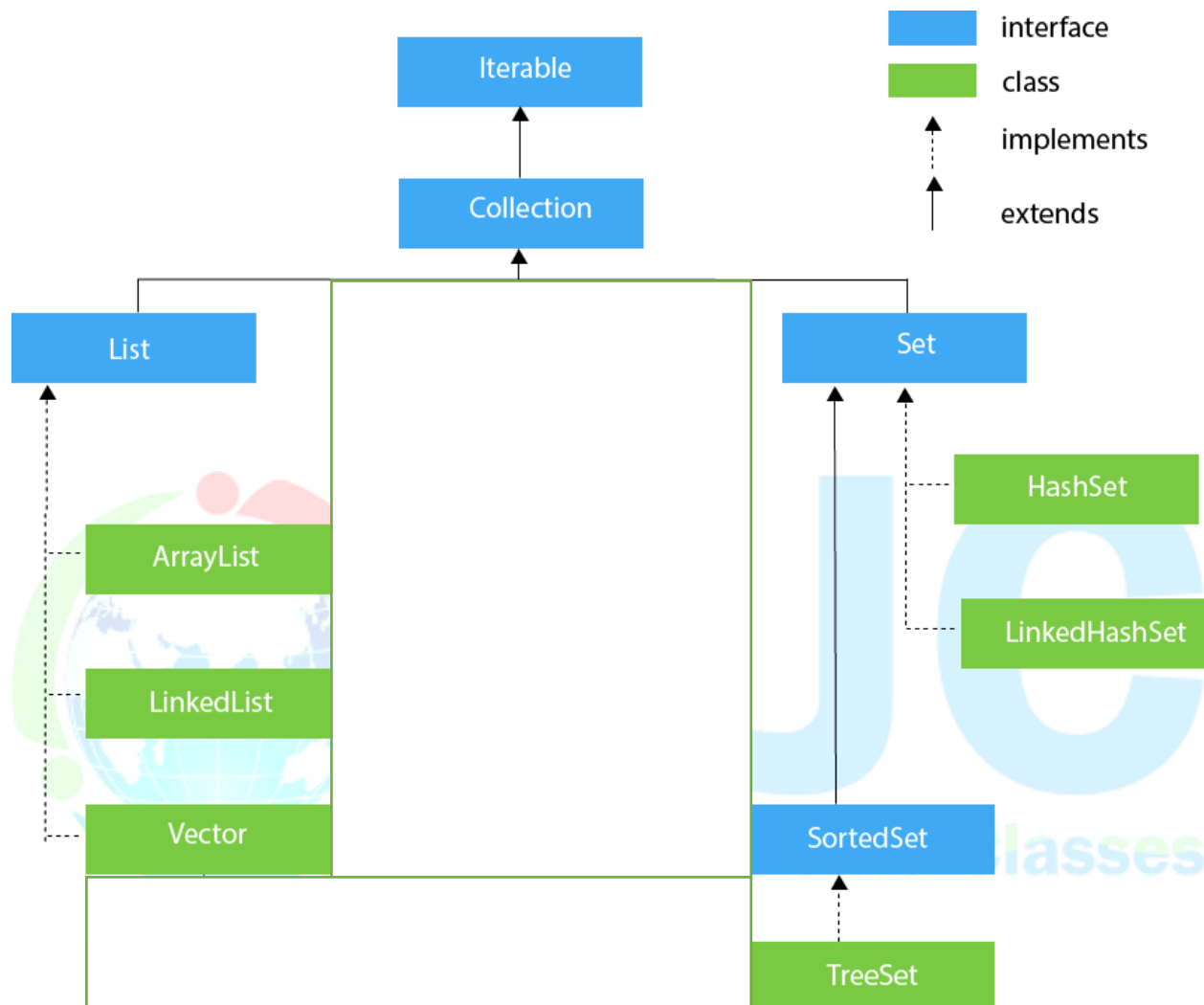
- Java collection framework was introduced from java 1.2 version, before that java had some ad hoc classes like vector, hash tables etc to store and handle data.
- These classes were useful but did not have the same interfaces, i.e, the technique of storing data was different for different classes. That is why collection framework was introduced in java 1.2 to make these classes unified.
- A collection framework in java is a container object that is used for storing multiple homogeneous and heterogeneous, duplicate and unique elements without any size limitations.

Hierarchy of Collection Framework:

The complete exception hierarchy is given in below diagram -



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List Interface from Collection:

What is List?

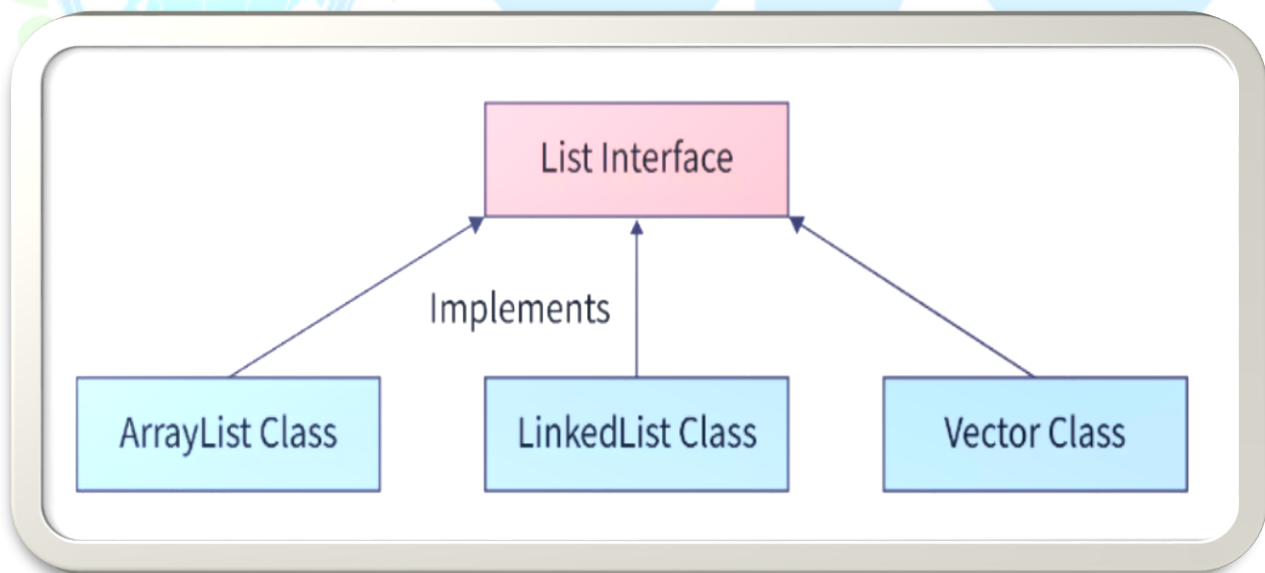


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- List is index-based data structure.
- A list is used to store **ordered collection** of data and it may contain duplicates and null.
- **Ordered Collection** means the order in which the elements are being inserted and they contain specific value.
- The elements present, can be accessed or inserted by their position in the list using zero-based indexing.
- The list interface is implemented by ArrayList, LinkedList and Vectors classes.
- List interface is the child interface of collection.
- List interface is found in **java.util** package.





Syntax of List Interface:

```
List<Obj> list = new ArrayList<Obj>();
```

Where: - List → Interface
<Obj> → Generic (inside angular bracket)
list → Reference variable / object
new → Keyword
ArrayList → Implemented Class
() → Parentheses

Note – Obj is the type of the object to be stored in list.

Operations in a Java List Interface:

Since list is an interface, it can be used only with a class that implements this interface. Now, let's see how to perform a few frequently operations on the list-

- **Operation 1:** Adding elements to List class using **add()** method.
- **Operation 2:** Updating elements in List class using **set()** method.
- **Operation 3:** Searching for elements using **indexOf()** method.
- **Operation 4:** Removing elements using **remove()** method.
- **Operation 5:** Accessing elements in List class using **get()** method.



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What are Generics in Java?

Generics means parameterized types. The idea is to allow type (Integer, String .. etc. and user defined types) to be parameter to methods, classes and interfaces.

Using generics, it is possible to create classes that work with different data types. An entity such as class, interface or methods that operates on a parameterized type is a generic entity.

Java Generics is one of the most important feature introduced in Java 5.

Why Generics in Java?

The Object is the superclass of all other classes, and Object reference can refer to any object. These features lack type safety. **Generics add that type of safety feature.**

What is an Iterator in Java?

- Iterator in java is used in the Collection Framework to retrieve elements one by one.



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- It is a universal iterator as we can apply it to any collection object.
- By using iterator, we can perform both read and remove operations.
- The iterator is the only cursor available for the entire collection framework.
- An iterator object can be created by calling the `iterator()` method present in the collection interface.

Syntax of an Iterator:

```
Iterator itr = c.iterator();
```

Note → Here “c” is any Collection object. itr is of type iterator interface and refers to “c”.

Methods of Iterator Interface:

The iterator interface defines some methods as listed below -

1. **hasNext()**: Returns true if the iteration has more elements.
2. **next()**: Returns the next element in the iteration.